# Data Science Tools: Languages, Libraries & APIs
(MODULE 1)

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# Data Science Languages Overview

**In Data Science Programing language are for:**

- Data manipulation & analysis
- Statistical modeling & ML
- Data visualization
- Scientific computing

**Why the right choice matter:**

- Faster result
- Better communication and collaboration due to popularity
- Wider job opportunity
- Some task are easier

# Python - “The Generalist”

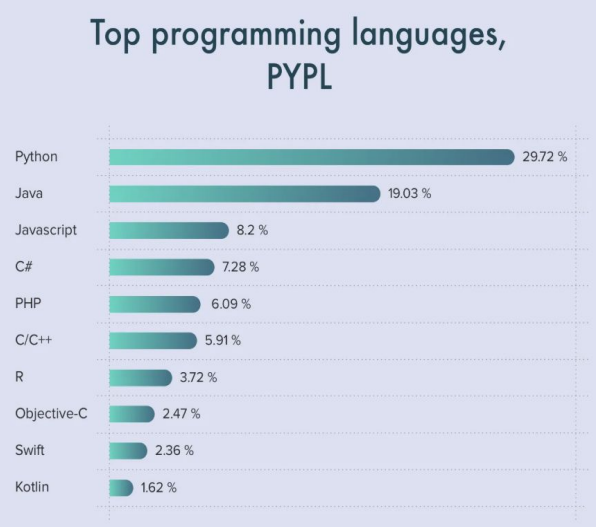
**Python**(1991) created for simplicity and readability. Gained popularity due to its powerful libraries and industry support.

<u>Best for:</u>

- Machine learning & AI
- Web development + Data
- Production system

<u>Many Libraries:</u> Pandas (data manipulation), NumPy (numerical), TensorFlow (machine learning),PyTorch (deep learning)

Top 7 Companies That Use Python Are Making A Mark In 2026




The PYPL index are base on language tutorials are searched on Google.

# R - “build for datas”

**R**(1993) specifically created for statistical computing. Gained popularity due to its powerful data analysis and visualization tools.

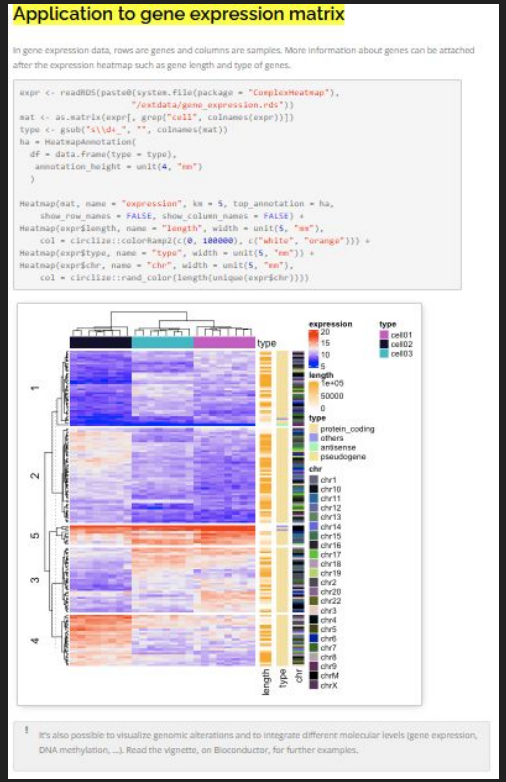
<u>Best for:</u>

- Statistical Analysis
- Data Visualization
- Academic Research

<u>Key libs:</u> ggplot2 (visualization), dplyr (data manipulation)

**Some applications of R**

1. UK Office for National Statistics - Produces official government statistics with R
2. Cancer Research (University of Salzburg) - Analyzes tumor genetics using R
3. Spotify - R is used by developers to analyze user listening habits
4. Netflix - Stock price prediction using machine learning in R
5. Google Play Store - Review analysis and sentiment tracking with R




R is used in cancer research to visualize gene expression patterns across different cell samples

# Matlab - "Numerical Computing Perfected"

**MATLAB**(1984) created for numerical computing. It is widely used in the fields of engineering, physics, and mathematical simulations.

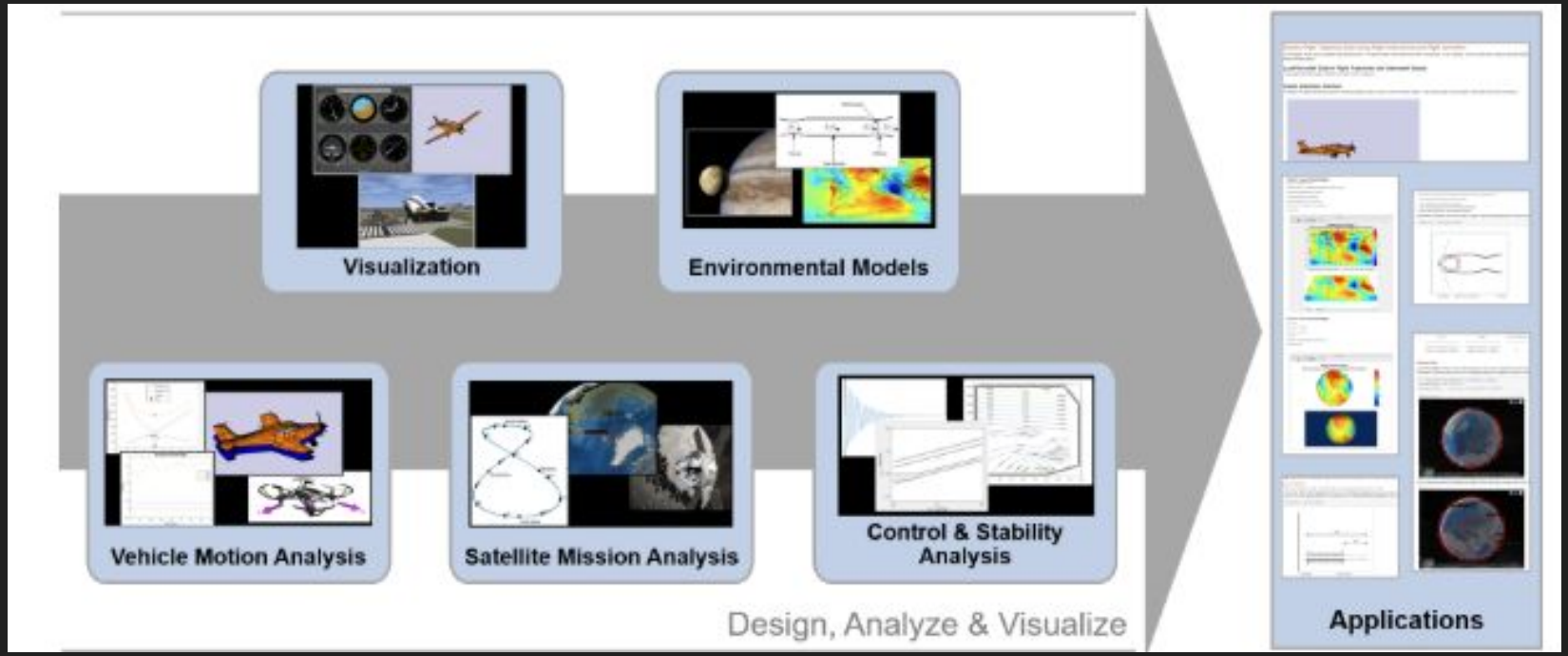
<u>Best For:</u>

- Numerical Computing
- Engineering Simulations
- Signal & Image Processing

<u>Key toolbox:</u> simulink, signal processing, control system

**Wide Known Companies that uses Matlab**

1. NASA: Utilizes MATLAB for data analysis, simulation, and modeling in various aerospace projects.
2. Tesla: Employs MATLAB for developing algorithms and simulations related to electric vehicle technology and autonomous driving systems.
3. Intel: For modeling circuit behavior and performance characterization.
4. Apple: In various engineering and design applications.




“**Aerospace Toolbox** provides standards-based tools and functions for analyzing the motion, mission, and environment of aerospace vehicles.” - Matlab Help Center

# Web Scraping Libraries

**Web scraping libraries** are tools that help you automatically extract data from websites.

Process: Website URL → Download HTML → Parse Structure → Extract Data → Save Results

| **Libraries** | Beautifulsoup | Scrapy | Selenium |
|---|---|---|---|
| **How does it work** | 1. You give it HTML<br>2. It converts HTML into a "parse tree"<br>3. You navigate the tree like a family tree<br>4. You search for specific elements by tag name, class, id, etc.<br>5. It returns the text/attributes you want | 1. You define "Spider" classes, with start URL, rules<br>2. Scrapy manages multiple requests.<br>3. It follows links automatically based on your rules<br>4. It can pause/resume, handle errors.<br>5. Built-in exporters | 1. Launches a REAL browser.<br>2. Browser renders JavaScript, loads dynamic content<br>3. You send commands like "click button"<br>4. Browser executes as if is a human<br>5. You extract data from the fully-loaded page |
| **Difficulty** | beginner | intermediate | advanced |

# APIs - RESTful vs GraphQL

**APIs**, Application Programming Interfaces, are invisible backbone enable applications and systems to communicate and share data efficiently.

Process: Your App → [HTTP Request] → API Server → Database (also works inverted)

**Company application:**
REST: Twitter API, Google Maps, Stripe
GraphQL: GitHub, Shopify, Facebook (they invented it!)

| APIs | RESTful | GraphQL |
|---|---|---|
| Ex / diff | `GET /pizzas`<br>`GET /pizzas/123`<br>`POST /orders`<br>`{`<br>`"pizzaId": 123,`<br>`"Address":`<br>`"123 GFG Office"`<br>`}`<br>(predefined pizza and toppings) | `query {`<br>`pizza(id: 123)`<br>`{`<br>`size`<br>`crust`<br>`toppings {`<br>`name`<br>`}`<br>`}`<br>`}` |
| Pro & Cons | Pro: Simple, caching<br>Con: Over/under-fetching | PRO: Ask for exactly what you need<br>CON: Complex query language |

# Comparison Table

| Feature | Python | R | MATLAB |
|---|---|---|---|
| **Ease of Learning** | Easy | Moderate | Moderate |
| **Scalability** | Highly scalable for large systems and big data | Moderate | Moderate |
| **Library Support** | Many Libraries(NumPy, pandas, TensorFlow) | Strong statistical package library | Moderate(Toolbox-based) |
| **Integration** | Excellent with web apps, cloud, APIs | Limited integration(Works with fewer tools) | Mostly standalone(Works mainly by itself) |
| **Use Case Strength** | High (Machine learning, automation, data science) | High (Statistical modeling, research analysis) | High (Numerical computing, simulations) |
| **Memory Efficiency** | Moderate | Moderate | High |
| **Flexibility (Cost)** | Free | Free | Paid |

# Pros & Cons

## Python

### Pros

- Highly versatile
- Human readable
- Free(Open source)
- Strong community
- Integration

### Cons

- Managing all dependencies
- Slower than compiled languages*

## MatLab

### Pros

- Specializes in linear algebra, engineering computations
- Readable syntax
- Great visualizations

### Cons

- Requires extensive prior knowledge
- Expensive!
- Not as versatile

## R

### Pros

- Specializes in statistical data analysis*
- Data mining
- Free(Open source)
- Great visualization

### Cons

- Unusual syntax
- Smaller ecosystem despite being open source
- Slower computation

# Reference

- 2026 programming language ranking
- Python popularity
- Heatmap in R
- Companies that uses python
- matlab applications
- Matlab help center
- Web scrap tutor
- MatLab vs Python vs R
- API
- RESTful vs GraphQL

# Q & A